

PRACTICAL : 8

Aim. Write a R program to create a complex number.

Problem statement :

1. Creating a numeric data type
2. Creating an integer
3. Creating a complex number
4. Converting from one type to another

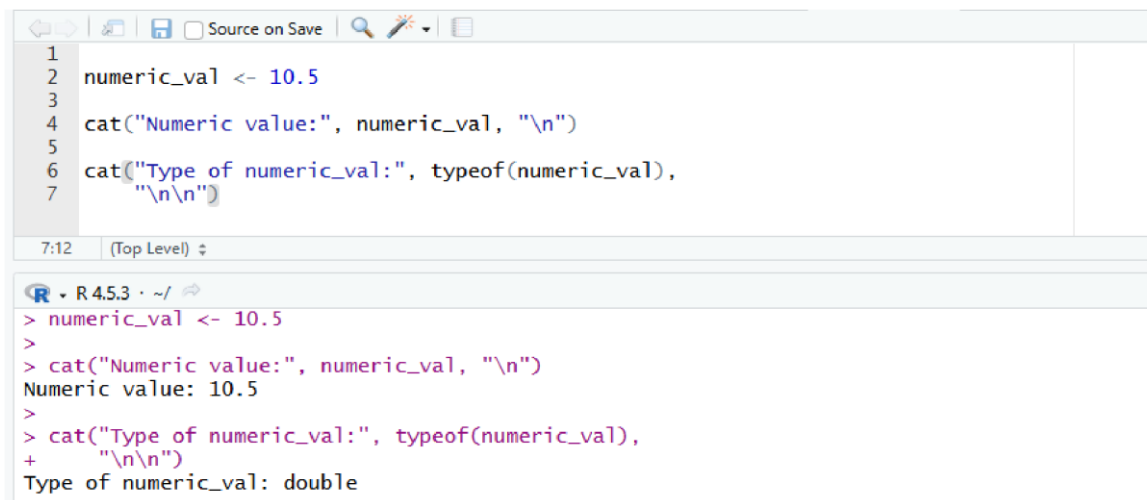
Theory :

In R, complex numbers are a basic data type used to store values with both real and imaginary components.

They are written in the form $Z = a + bi$, where i represents the imaginary unit.

Code :

1. Creating a numeric data type



```
1 numeric_val <- 10.5
2
3
4 cat("Numeric value:", numeric_val, "\n")
5
6 cat("Type of numeric_val:", typeof(numeric_val),
7     "\n\n")
```

7:12 (Top Level) ↕

R 4.5.3 · ~/

```
> numeric_val <- 10.5
>
> cat("Numeric value:", numeric_val, "\n")
Numeric value: 10.5
>
> cat("Type of numeric_val:", typeof(numeric_val),
+     "\n\n")
Type of numeric_val: double
```

2. Creating an integer

```
9
10 integer_val <- 20L
11
12 cat("Integer value:", integer_val, "\n")
13
14 cat("Type of integer_val:", typeof(integer_val),
15     "\n\n")
16
```

9:1 (Top Level) ⚡

R 4.5.3 · ~/

```
>
> integer_val <- 20L
>
> cat("Integer value:", integer_val, "\n")
Integer value: 20
>
> cat("Type of integer_val:", typeof(integer_val),
+     "\n\n")
Type of integer_val: integer
```

3. Creating a complex number

```
18 complex_val <- 4 + 5i
19
20 cat("Complex number:", complex_val, "\n")
21
22 cat("Type of complex_val:",
23     typeof(complex_val), "\n\n")
24
25
```

17:1 (Top Level) ⚡

R 4.5.3 · ~/

```
>
> complex_val <- 4 + 5i
>
> cat("Complex number:", complex_val, "\n")
Complex number: 4+5i
>
> cat("Type of complex_val:",
+     typeof(complex_val), "\n\n")
Type of complex_val: complex
```

4. Converting from one type to another

```
1 # Convert numeric to integer
2
3 converted_integer <- as.integer(numeric_val)
4 cat("Converted numeric to integer:", converted_integer, "\n")
5 cat("Type of converted_integer:", typeof(converted_integer), "\n\n")
6
7 # Convert integer to complex
8
9 converted_complex <- as.complex(integer_val)
10 cat("Converted integer to complex:", converted_complex, "\n")
11 cat("Type of converted_complex:", typeof(converted_complex), "\n\n")
12
13 # Convert complex to numeric (real part only)
14
15 converted_numeric <- as.numeric(complex_val)
16 cat("Converted complex to numeric (real part):", converted_numeric, "\n")
17 cat("Type of converted_numeric:", typeof(converted_numeric), "\n")
18
19 17:66 (Top Level) ±
```

```
R 4.5.3 ~ /
> converted_integer <- as.integer(numeric_val)
> cat("Converted numeric to integer:", converted_integer, "\n")
Converted numeric to integer: 10
> cat("Type of converted_integer:", typeof(converted_integer), "\n\n")
Type of converted_integer: integer
>
> # Convert integer to complex
>
> converted_complex <- as.complex(integer_val)
> cat("Converted integer to complex:", converted_complex, "\n")
Converted integer to complex: 20+0i
> cat("Type of converted_complex:", typeof(converted_complex), "\n\n")
Type of converted_complex: complex
>
> # Convert complex to numeric (real part only)
>
> converted_numeric <- as.numeric(complex_val)
Warning message:
imaginary parts discarded in coercion
> cat("Converted complex to numeric (real part):", converted_numeric, "\n")
Converted complex to numeric (real part): 4
> cat("Type of converted_numeric:", typeof(converted_numeric), "\n")
Type of converted_numeric: double
```

Conclusion : R program to create a complex number is executed successfully .